

Take-Home



AfterQuery Research Scientist (Post-Training) Take Home:

Task Description

The core task is to select a popular, open-source Reinforcement Learning framework and use it to create an **RLVR (Reinforcement Learning from Virtual Reward) environment**. This environment will be designed for improving a model's performance on a specific benchmark. This should not take longer than 4-8 hours.

1. Framework Selection

The candidate must select **and justify** the use of a popular, actively maintained RL framework. (Why did you pick it?)

2. Benchmark Selection

The candidate must choose **one** of the following two benchmarks as the target for model improvement:

- **terminalbench2**
- **tau2 bench**

3. Environment Creation

Using the chosen RL framework, the candidate must outline the design and implementation of the RLVR environment.

Key Deliverables:

1. **State/Observation Space Definition:** Clearly define the observation space that the RL agent receives from the environment (e.g., token sequences, embeddings, prompt structure).
2. **Action Space Definition:** Clearly define the set of actions the RL agent can take (e.g., generating the next token, selecting a specific output, applying a correction).
3. **Reward Function Design (minimum 2):** Detail the virtual reward functions, which should incentivize the model to perform better on the chosen benchmark's task. This function is critical for the RLVR approach.

4. Model Selection and Training Plan

The candidate must propose a suitable base model and a detailed training plan for the RL phase.

Component	Description
Base Model Selection	Specify the LLM that would be fine-tuned (e.g., Llama-2-7b, Mistral-7B). Justify the choice based on the benchmark task.
RL Algorithm	Identify the specific RL algorithm to be used (e.g., PPO, GRPO, etc). Justify why this algorithm is appropriate for the defined state/action/reward space.
Dataset Size / Configuration	Num samples, curriculum, etc.
Hyperparameters	Define the essential training hyperparameters.
	- Learning Rate:
	- Batch Size:
	- Number of Training Steps/Epochs:
	- Optimizer:
	- etc.

Component	Description
Metrics	Define the key metrics to track during training (e.g., reward per episode, benchmark accuracy).

5. Deliverables

The final submission should be a document containing:

- A detailed write-up of the environment design and training plan, following the sections above.
- A link to the proposed environment code repository

Stretch Goals (Optional)

Candidates who complete the core task may pursue the following stretch goals to further demonstrate their capabilities:

- **Synthetic Task Generation:** Synthetically generate one novel task that conforms to the format and constraints of the chosen benchmark (`terminalbench2` or `tau2 bench`).
- **Dual Environment Creation:** Outline or prototype the RLVR environment for **both** `terminalbench2` and `tau2 bench`.
- **Scaling:** How would you change the environment + your hyper parameters for a larger or smaller model?

Next Steps

Send your code (**PRIVATE GIT REPO**) + write up to michael@afterquery.com for review.